NORTHEASTERN UNIVERSITY

IE7615 Deep Learning

Detecting 73 Everyday Objects in Multi-object Images with YOLO

Discriminative Project: Milestone 2 Report

Submitted by: Group 7

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Abstract

We present a detector that reports the Object ID and bounding box of every object in images composed from a 73-class dataset of single-object photographs collected by the course. The dataset carries class labels but no boxes, so the central difficulty is producing box labels that are accurate enough to train on and trustworthy enough to evaluate against. We address it with a consensus annotation pipeline: an open-vocabulary detector prompted with each object's name (OWLv2), an image-guided variant of the same detector, and a YOLO26-s model self-trained on the training split vote on every box, and a person checks every test box and a random sample of training boxes. Auditing all boxes of each class side by side revealed classes whose boxes followed different definitions of the object (one shoe or the pair, the tissue paper or the tissue box, the watch case or the whole watch), and correcting these definitions produced the largest accuracy gains of the project. From the labelled photos we synthesise 2x2 to 5x5 grids and single-photo images and fine-tune COCO-pretrained YOLO26-s and YOLOv8-s detectors at an input size of 1120 pixels. The selected model reaches **mAP@0.5 = 0.947**, **mAP@0.5:0.95 = 0.875**, **precision = 0.931** and **recall = 0.918** on grids built exclusively from held-out test photos with human-verified boxes; it detects single photos without error and assigns a wrong Object ID to 16 of 1,041 objects. Three alternative pipelines explored within the group, each with automatic labels only, were cross-evaluated on the same human-checked objects with the same code, eight runs in total: the selected model leads all three alternatives by 6 to 14 points of mAP@0.5 and 9 to 15 points of mAP@0.5:0.95, and one alternative that scored 0.947 on its own automatically labelled test set reached only 0.779 on unseen human-checked objects. Remaining errors concentrate in groups of visually similar objects, in very small objects and in degraded images.

Keywords: object detection, YOLO, open-vocabulary detection, pseudo-labelling, label quality, synthetic composites, cross-evaluation

1. Introduction

The course dataset contains photographs of 73 everyday objects, each photographed by one student in many settings. Milestone 1 established that a CNN classifier can name the object in such a photo with about 99% accuracy. Milestone 2 asks for more: given an image containing *several* objects, the model must report the Object ID **and** the location of each one. This is an object-detection task, and it differs from classification in three ways that shaped our work.

1. **No box labels exist.** Every detector needs a bounding box for every training object, and every evaluation needs a correct box for every test object. With about 7,000 photos, drawing all boxes by hand was impractical, yet automatic boxes cannot simply be trusted: a single wrong box in the test set makes a correct prediction look wrong.
2. **Objects must be located as well as named.** Many photos show their object small, far away or in clutter, and several students photographed visually similar objects (three watches, two wallets, four kinds of rackets, two power banks).

3. **Multi-object images must be synthesised.** The milestone specifies grids of 224 x 224 photos placed side by side, from 2x2 (4 objects) to 5x5 (25 objects), so a detector must cope with a six-fold range of object counts and image sizes.

Contributions. Our work makes six contributions:

- A **consensus annotation pipeline** that combines two automatic box sources with a self-trained second opinion and human review, with a **measured** accuracy of 98.9% for automatically accepted boxes (Section 4).
- A **class-level label audit** that found and corrected inconsistent definitions of five objects, raising the worst corrected class from AP@0.5 = 0.30 to 0.995 (Section 4.6).
- A **grid synthesis procedure** that follows the milestone specification exactly, uses every validation and test photo exactly once, and provably keeps every source photo in one split (Section 5).
- A **fine-tuned YOLO26-s detector** selected and post-processed on validation data only, evaluated once on human-verified test boxes (Sections 6 to 8).
- A **leakage-controlled cross-evaluation** of three alternative pipelines on our human-checked test set, showing how strongly automatically labelled test sets overstate accuracy (Section 9).
- An **honest error analysis** including the effect of grid size, look-alike objects, single photos and image degradation, together with the threats to validity of our protocol (Section 10).

2. Related Work

Single-stage detectors. YOLO [1] frames detection as one regression from image to boxes and class scores, which makes it fast enough to train repeatedly on a laptop GPU. Later versions added stronger backbones and training-time augmentation such as mosaic [2]. We use two Ultralytics models: YOLOv8 [3], the model named in the brief, and YOLO26 [4], the newest release of the same library. Both are pretrained on COCO [5], which supplies general notions of "objectness" that transfer to new classes.

Open-vocabulary detection. OWL-ViT [6] and its successor OWLv2 [7] pair a vision transformer with a text encoder, so they can find objects described by an arbitrary text prompt, and can also be conditioned on an example image instead of text. Grounding DINO [8] follows a similar idea. Such detectors are a natural source of boxes for a new dataset, but their outputs are noisy for unusual objects and object names.

Pseudo-labelling and label quality. Training a model on its own confident predictions, known as pseudo-labelling [9], is a standard way to exploit unlabelled data, but errors in the pseudo-labels propagate. Label errors also matter at evaluation time: Northcutt et al. [10] showed that errors in widely used *test* sets are common enough to change which model appears best. This motivated our decision to verify every test box by hand rather than estimate test quality.

Synthetic composites. Pasting object instances onto new backgrounds is an effective way to create detection training data [11, 12]. Our grids are a structured instance of this idea, prescribed by the milestone: photos are placed whole, side by side, so each object keeps its original background.

3. Dataset

3.1 Source and cleaning

We reuse the cleaned dataset and split from Milestone 1, so both milestones rest on identical data decisions. Starting from 7,343 files, 385 background-only photos and exact duplicates were removed, leaving **6,942 single-object photos** of 73 classes (89 to 106 per class). Duplicates were removed **before** splitting, so no copy of a photo can appear in two splits.

3.2 Splits and leakage control

The split is 70/15/15, stratified by Object ID with a fixed seed: **4,859 training, 1,041 validation and 1,042 test photos.** Every multi-object image of a split is built only from that split's photos, which guarantees that no source photo appears in more than one split; an automatic check confirms this for every generated image (Section 5.5).

3.3 Known dataset issues

Table 1. Known issues of the shared dataset and how they were handled.

Issue	Handling
Exact duplicate photos	Removed before splitting (Milestone 1)
One OBJ003 photo not 224 x 224	Every photo is resized to 224 x 224 (squashed, never cropped)
OBJ054_017 is a phone screenshot, not the object	Excluded, as instructed
Five OBJ004 photos show only the background	Excluded as not photos of the object
Many photos show the object small, far away or in clutter	Kept; a photo is left out only if no reliable box can be made (Section 4.7)

4. Bounding-Box Annotation

4.1 Design principles

Three principles guided the annotation pipeline. **(i) Independence:** a box is trusted only when methods that fail in different ways agree. **(ii) Human ground truth for evaluation:** every test box is checked by a person, because test labels define the reported numbers. **(iii) Measurement over assumption:** the accuracy of automatically accepted boxes is measured on human-checked samples, not assumed. Figure 1 summarises the pipeline.

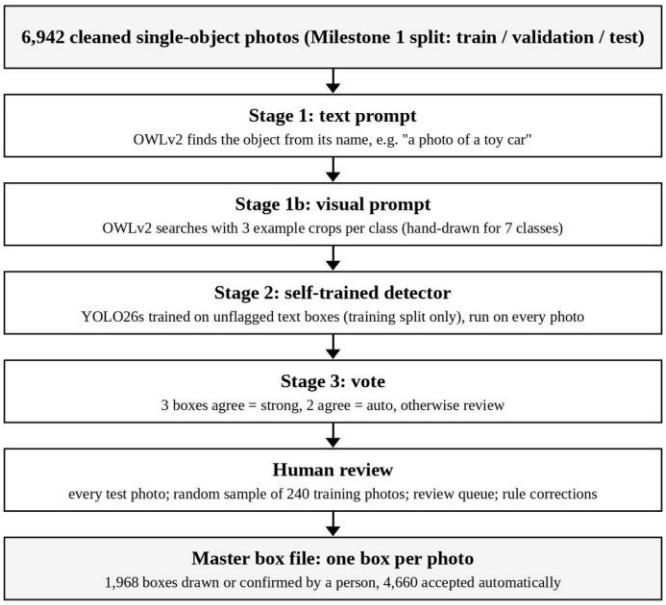


Figure 1. The consensus annotation pipeline. Two automatic box sources and a self-trained second opinion vote on every box; a person checks every test box and the boxes the vote cannot resolve.

4.2 Automatic box sources

Text prompt. OWLv2 [7] was prompted with each photo's object name from the Object List (for example "a photo of a toy car"), and the highest-scoring box was kept. Boxes with low confidence, very small boxes and boxes covering almost the whole photo were flagged. Of 6,936 photos, 5,570 (80.3%) received an unflagged box; the median confidence was 0.52.

Visual prompt. In its image-guided mode, OWLv2 searched each photo for three example crops of the same object, taken from training photos. For seven classes whose automatic examples were wrong or missing, the examples were drawn by hand (OBJ004, OBJ010, OBJ011, OBJ039, OBJ040, OBJ043, OBJ048). We reimplemented the search so that the examples of a class are embedded once rather than with every photo, and verified on 12 photos that the faster version scores identically to the library method (score-pattern correlation of at least 0.9987).

A negative finding. Plotting every photo of a class with all its boxes showed that the visual-prompt boxes were frequently large regions of floor, bed or table: text and visual boxes agreed (overlap of at least 0.7) on only 47.9% of photos, with a median overlap of 0.66. The example crops themselves contain background, so large background regions score as similar. We therefore removed the visual prompt from the role of gatekeeper and kept it only as one vote.

4.3 Self-trained second opinion and consensus vote

A YOLO26-s model was trained for 40 epochs on the 3,915 unflagged text-prompt boxes of the **training split only** (validation mAP@0.5 = 0.950 on single photos) and then run on every photo. Because it never saw a validation or test photo, its boxes on those splits are an independent second opinion. Each photo was then assigned one of four decisions (Table 2):

Table 2. Consensus decisions over all 6,936 photos.

Decision	Rule	Photos
Strong	All three boxes agree	3,172 (45.7%)
Auto	Two boxes agree	2,935 (42.3%)
Review	No two boxes agree	794 (11.4%)
Review label	The self-trained model confidently sees a *different* Object ID	35 (0.5%)

4.4 Human verification

All reviewing was done in a purpose-built local tool with two modes: a verification grid of 24 photos with their boxes, in which wrong boxes are clicked, and a correction view of one photo, in which a suggested box is chosen or a new one drawn. Every action is saved to disk immediately, so a session can be stopped and resumed. Table 3 lists the human work.

Table 3. Human review.

Review step	Photos	Outcome
Every test photo	1,041	885 confirmed, 156 corrected
Random sample of automatically accepted training boxes	240	233 correct, 7 flagged
Classes without usable automatic boxes (OBJ040, OBJ043)	about 190	boxed by hand
Eight classes that would otherwise lose more than 20% of their photos	review queue	hard photos boxed by hand, so they remain in the dataset
Label-definition corrections (Section 4.6)	5 classes, all splits	re-verified and corrected
Total boxes drawn or confirmed by a person	**1,968**	

4.5 Measured label quality

Because every test box was reviewed, the review doubles as a measurement of how accurate the automatically accepted boxes are (Table 4).

Table 4. Accuracy of automatically accepted boxes, measured against human review.

Measurement	Photos checked	Correct
Auto-accepted boxes on the test split (all of them)	873	863 (**98.9%**)
of which decision "strong"		99.6%
of which decision "auto"		98.0%
Random sample of auto-accepted training boxes	240	233 (**97.1%**)
Photos sent to review: median overlap of the automatic box with the human box		0.23

The vote therefore separates good from bad automatic boxes sharply: accepted boxes are almost always correct, while the boxes it rejected rarely matched what a person drew. The training-split figure is slightly lower than the test figure because the self-trained model learned from the training text boxes and tends to agree with them, which makes agreement weaker evidence on that split. The remaining 4,660 training and validation boxes were accepted automatically on the basis of these measurements.

4.6 Label-definition audit

Agreement between methods guarantees that boxes are *consistent with each other*, not that they follow a single definition of the object. Plotting all photos of a class with their final boxes revealed five classes whose boxes followed two definitions (Table 5, Figure 2). Inconsistent labels give the detector two incompatible targets for one class, so each case was given one explicit rule and corrected by hand in every split.

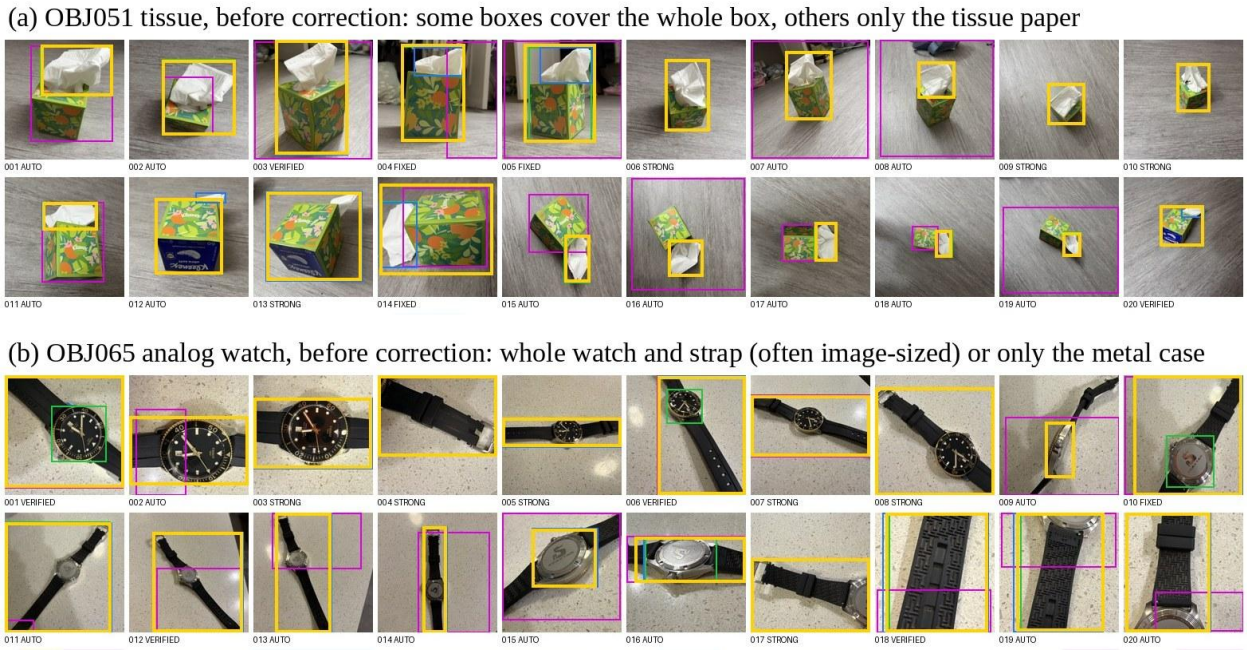


Figure 2. Inconsistent object definitions found by the class-level audit (yellow: box kept before correction). (a) OBJ051: the whole tissue box or only the tissue paper. (b) OBJ065: the whole watch and strap, often as large as the image, or only the metal case.

Table 5. Label definitions found inconsistent, the rule adopted, and the effect on test AP@0.5.

Object ID	Inconsistency found	Rule adopted	AP@0.5 before	AP@0.5 after
OBJ010 Nike sneakers	Box on one shoe of the pair, sometimes the left and sometimes the right	One box around **both shoes**	inconsistent	0.995
OBJ051 Tissue	Box often on the tissue paper only: the prompt "tissue" was read as the paper	One box around the **whole tissue box, including the tissue**	0.30	**0.995**
OBJ065 Analog watch	Some boxes covered the whole watch and strap; because the strap runs out of the image, these boxes were as large as the image. Others covered only the metal watch case in the middle	The **whole visible watch, case and strap**, up to the image edge where the strap leaves the frame	0.59	**0.81**
OBJ048 Wireless power bank	Several boxes on small nearby objects	The silver power bank only	0.71	**0.95**
OBJ034 Table-tennis racket	A few boxes on a tennis racket lying next to the paddle	The small paddle only	0.67	0.59

The two other watch classes, OBJ025 and OBJ046, already followed the case-and-strap rule, so all three watches share one definition; the box sheets of further classes with an ambiguous extent or many review cases (purse, sleep mask, guitar, card wallet, power bank and mouth-guard case) were checked and found consistent. OBJ034 did not improve after correction, which indicates that its remaining errors stem from its visual similarity to the other rackets rather than from labels (Section 8.5).

4.7 Photos left out

A photo was left out only when no reliable box could be made: **314 of 6,942 photos (4.5%)**, of which 6 are not photos of the object (Table 1) and 308 had no agreeing automatic box and were not reviewed. No Object ID loses more than 20% of its photos (the highest is OBJ059 at exactly 20.0%), 27 lose none, and small, distant and cluttered photos were kept wherever their object could be labelled. Appendix A lists every Object ID.

5. Multi-object Image Synthesis

5.1 Grid construction

Multi-object images are built by placing 224 x 224 photos side by side in the five sizes specified for the milestone (Table 6, Figure 3). Each photo's box is shifted by the position of its cell, so every grid label is exact by construction; the generator was verified by placing coloured squares at known positions and confirming that every one of over 1,000 generated boxes lies on its square, including after flips.

Table 6. Grid sizes.

Grid	Objects per image	Image size (pixels)
2x2	4	448 x 448
3x3	9	672 x 672
4x4	16	896 x 896
5x4	20	1120 x 896
5x5	25	1120 x 1120

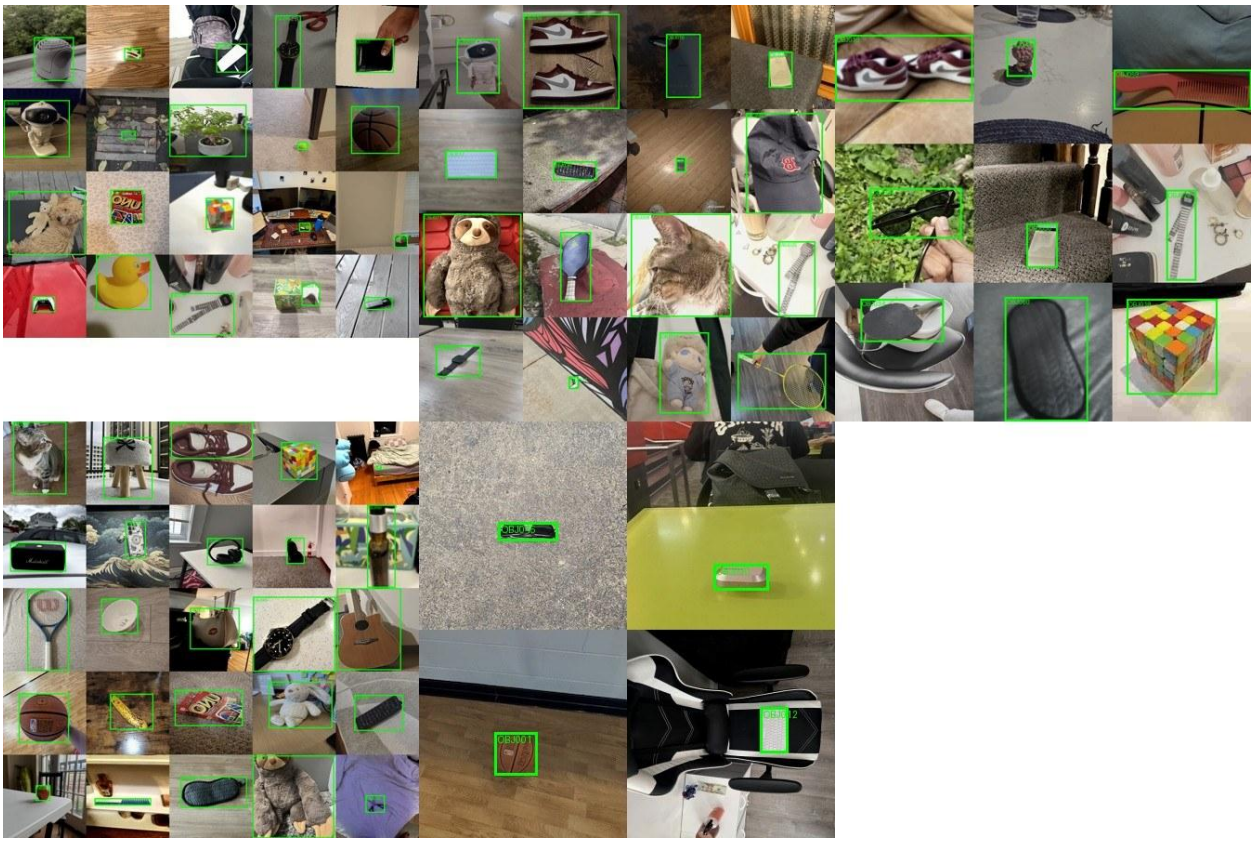


Figure 3. Training images with their ground-truth boxes (green), one example of each grid size.

5.2 Training images

The training set contains **1,500 images**: 85% grids, with an equal share of the five sizes, and 15% single photos, so that single images from the dataset are also detected. Photos are drawn so that every Object ID appears equally often and every photo is reused before any is used again; within an image, Object IDs are distinct.

5.3 Look-alike groups

Students photographed several visually similar objects. We defined ten groups from the **names** in the Object List, not from test results, for example *watch, iWatch, analog watch*; *card wallet, men's wallet*; *tennis racket, table-tennis racket, badminton racket, pickleball bat*; *wireless power bank, power bank*; *sunglasses,*

wayfarer sunglasses, swimming goggles. Half of the multi-object training images contain one complete group side by side, so the detector repeatedly sees the small differences between its members in the same image.

5.4 Augmentation

Each training photo is horizontally flipped with probability 0.5, its brightness, contrast and saturation are varied by up to 20%, and 20% of photos receive one real-photo degradation: blur, a reduction to 30 to 55% resolution, heavy JPEG compression or darkening. Milestone 1 identified low-quality photos as the hardest condition, which motivated this choice. Boxes are transformed together with the photos.

5.5 Evaluation sets

Validation and test images use **every usable photo of their split exactly once**, without augmentation, so each object is counted once. The **main test set** has 72 grids (16 2x2 and 14 of each other size) built from the 1,041 test photos. A **second test set** of 84 images, built from the same test photos but arranged as single photos and the five grids, measures every input type that can be given. An automatic check confirms, for all evaluation sets, that no source photo appears in more than one split, that every photo is placed exactly once and that every label line is valid.

6. Detection Models and Training

6.1 Models and settings

We fine-tune two COCO-pretrained small detectors, **YOLOv8-s** [3] and **YOLO26-s** [4]. The input size is **1120 pixels**, the size of the largest grid, so that photos in 5x4 and 5x5 grids keep their native 224 pixels instead of being shrunk. Table 7 lists the remaining training settings.

Table 7. Training settings.

Setting	Value
Input size	1120
Batch size	4, to fit an 8 GB laptop GPU (RTX 5070) at this input size
Epochs	up to 60, early stopping after 12 epochs without improvement
Learning-rate schedule	cosine
Mosaic augmentation	on half of the images, switched off for the last 10 epochs
Geometric augmentation	scale 25%, translation 10%, horizontal flip; no rotation, shear or mix-up
Training time	YOLO26-s: 60 epochs, 104 min; YOLOv8-s: stopped after 56 epochs, 70 min

6.2 Fine-tuning after data improvements

Instead of retraining from scratch after each data improvement, both models were fine-tuned from their trained weights for 15 epochs at a lower learning rate (0.002, no warm-up). The **first fine-tune** followed

the OBJ010 correction. The **second fine-tune**, of YOLO26-s, followed the addition of single photos and look-alike groups and the corrections of OBJ051, OBJ065, OBJ048 and OBJ034.

6.3 Model selection and post-processing

All five trained models were compared on **validation**, and the model with the highest validation mAP@0.5:0.95 was selected before the test set was evaluated. Two post-processing options were also chosen on validation only, each together with a confidence threshold, with "no post-processing" always among the candidates:

- **One box per grid cell.** Each 224 x 224 cell of a grid holds exactly one photo, so only the most confident box whose centre lies in a cell is kept. This removes duplicate boxes, such as two boxes on the two shoes of a pair, that an overlap-based rule cannot remove. The rule is applied only when the image dimensions are multiples of 224.
- **Natural-size inference.** A small input (a single photo or a small grid) is placed unscaled on a 1120 x 1120 canvas, so every photo is seen at the size used in training instead of being enlarged up to five times; boxes are mapped back to the input. On validation this scored higher than ordinary resizing (65.0% versus 57.5% fully correct images) and was selected.

7. Evaluation Protocol

Matching. A predicted box is a true positive when it overlaps an unmatched true box of the same Object ID with an intersection over union (IoU) of at least 0.5; predictions are matched in order of confidence.

Detection metrics. *Precision* is the share of predicted boxes that are true positives and *recall* the share of true boxes that are found. *AP* (average precision) summarises precision over all recall levels for one class, and *mAP@0.5* is its mean over the 73 classes, following the PASCAL VOC definition [13]. *mAP@0.5:0.95* averages mAP over IoU thresholds from 0.5 to 0.95, following COCO [5], and therefore also rewards tight boxes.

Task metrics. Because the task is to report *every* object of an image, we also count, per image, the objects found with the correct ID, the objects assigned a *wrong* ID, the missed objects and the extra boxes, and we report the share of **fully correct images**: images in which every object is found with the correct ID and no extra box is produced. This all-or-nothing measure is much stricter than the per-object metrics. Its 95% confidence interval is obtained by bootstrap resampling of images [14].

Protocol. Every choice (model, post-processing, confidence threshold, inference method) was made on validation data. The test set was used once per model, with boxes that were all verified by a person.

8. Results

8.1 Model comparison

Table 8. Validation and test results of all trained models.

Model	Val mAP@0.5	Val mAP@0.5:0.95	Test mAP@0.5	Test mAP@0.5:0.95	Test precision	Test recall
YOLOv8-s	0.940	0.897	0.910	0.846	0.914	0.890
YOLO26-s	0.943	0.907	0.921	0.858	0.922	0.884
YOLOv8-s, fine-tuned	0.952	0.906	0.932	0.859	0.925	0.894
YOLO26-s, fine-tuned	0.956	0.919	0.931	0.863	0.922	0.910
YOLO26-s, fine-tuned twice (selected)	**0.972**	**0.930**	**0.947**	**0.875**	**0.931**	**0.918**

The two architectures perform within about one point of each other, with YOLO26-s ahead on the stricter mAP@0.5:0.95. The larger differences come from the data: the first fine-tune adds 1.0 point of test mAP@0.5 to YOLO26-s and 2.2 points to YOLOv8-s, and the second, which combines label corrections with single photos and look-alike groups, adds a further 1.6 points. These gains combine data changes with additional training epochs, so they are not a clean ablation of any single change; per-class results (Section 8.5) attribute the largest share to the label corrections.

8.2 Results per grid size

Table 9 breaks the main test results down by grid size.

Table 9. Test results of the selected model per grid size (main test set; confidence threshold 0.1 and one box per cell, both chosen on validation).

Grid	Images	Recall	Precision	Wrong ID	Missed	Extra boxes	Fully correct
2x2	16	98.4%	96.8%	0	1	2	87.5%
3x3	14	89.7%	92.6%	3	10	6	35.7%
4x4	14	91.1%	92.7%	4	16	12	14.3%
5x4	14	91.8%	93.8%	5	18	12	21.4%
5x5	14	92.0%	95.8%	4	24	10	7.1%
All	**72**	**91.8%**	**94.3%**	**16**	**69**	**42**	**34.7%** (95% CI 23.6 to 45.8%)

Per object, the model finds 90 to 92% of objects with the correct ID at every grid size beyond 2x2, which shows that it scales to 25 objects per image. The share of fully correct images falls with grid size because the measure is all-or-nothing: with roughly 92% per object, the probability that all 25 objects of a 5x5 grid are right is about 0.92 raised to the power 25, or 12%, close to what we observe. Errors are dominated by **missed objects** (69), most of them very small or distant inside their photo; wrong Object IDs are rare (16 of 1,041 objects). Without post-processing, 29.2% of images are fully correct, so the one-box-per-cell rule adds 5.5 points, mainly by removing duplicate boxes.

8.3 Single photos and every input type

Table 10 extends the evaluation to every input type, from single photos to 5x5 grids.

Table 10. Results on the second test set (test photos only) with natural-size inference.

Input	Images	Recall	Precision	Fully correct
Single photo	14	**100%**	**100%**	**100%**
2x2	14	80.4%	86.5%	28.6%
3x3	14	91.3%	95.0%	35.7%
4x4	15	92.1%	94.0%	20.0%
5x4	14	91.8%	95.9%	14.3%
5x5	13	93.5%	95.3%	15.4%

mAP on this set, computed with ordinary resizing: mAP@0.5 = 0.941, mAP@0.5:0.95 = 0.866.

Every single photo is detected with the correct Object ID, a tight box and no extra box. The 2x2 grids score lower here than in the main test set (80.4% against 98.4% recall). Both sets contain only 14 to 16 images of this size, made from different photos, so per-size figures are uncertain by several points; the gap is consistent with this sampling variation, but we cannot rule out that natural-size inference helps single photos and large grids more than small grids. On this test set ordinary resizing would have produced slightly more fully correct images (39.3% versus 35.7%), the opposite of validation and within the same uncertainty.

8.4 Robustness to image degradation

On a degraded copy of the main test set (resolution reduced four-fold, brightness reduced by 40%, JPEG quality 20), mAP@0.5 falls from 0.947 to **0.428** and mAP@0.5:0.95 to 0.342. Low-quality images remain the hardest condition, as in Milestone 1. The first fine-tuned model scored 0.506 on the same degraded set, so the second fine-tune traded some robustness to heavy degradation for higher accuracy on normal images and single photos.

8.5 Results per Object ID

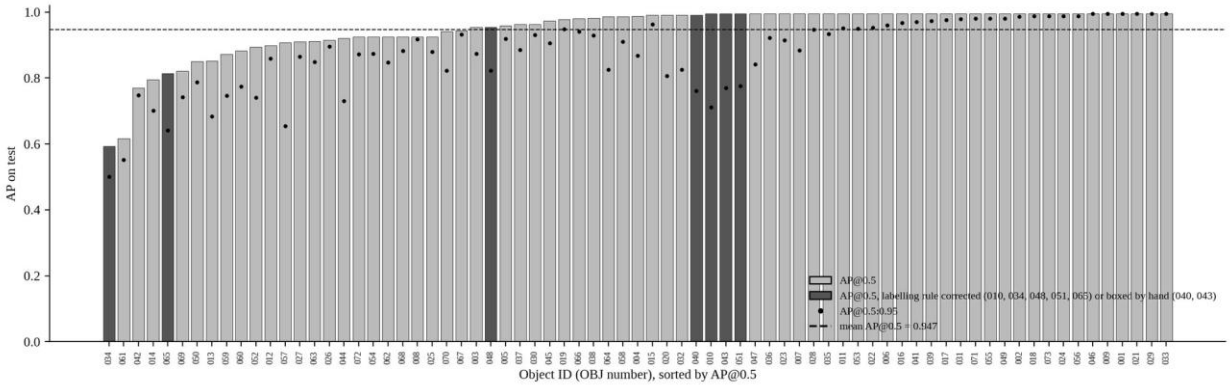


Figure 4. Test AP of the selected model for all 73 Object IDs, sorted by AP@0.5. Dark bars mark classes whose labelling rule was corrected or which were boxed by hand.

The mean AP@0.5 over the 73 classes is 0.947; **48 classes reach at least 0.95, 61 at least 0.90, and 4 fall below 0.80** (Figure 4). All four weakest classes are members of look-alike groups, and the confusion matrix (Figure 5) shows their errors as off-diagonal entries towards other members of their group. Table 11 lists the four weakest classes.

Table 11. The four weakest classes.

Object ID	AP@0.5	AP@0.5:0.95	Likely cause
OBJ034 Table-tennis racket	0.59	0.50	Four kinds of rackets; the paddle is often small or held in a hand
OBJ061 Card wallet	0.62	0.55	Resembles the men's wallet (OBJ067); often small in cluttered rooms
OBJ042 Mouth-guard case	0.77	0.75	Small object in cluttered scenes; shares photos with the tennis ball (OBJ027)
OBJ014 Sunglasses	0.79	0.70	Resembles the wayfarer sunglasses (OBJ045)

The classes labelled entirely by hand (OBJ040: 0.99; OBJ043: 0.995) and four of the five corrected classes perform well, which supports the conclusion that consistent labels matter more than the source of a label. With about 14 test photos per class, a single miss changes a class's AP by roughly 0.07, so differences of a few points between classes should not be over-interpreted.

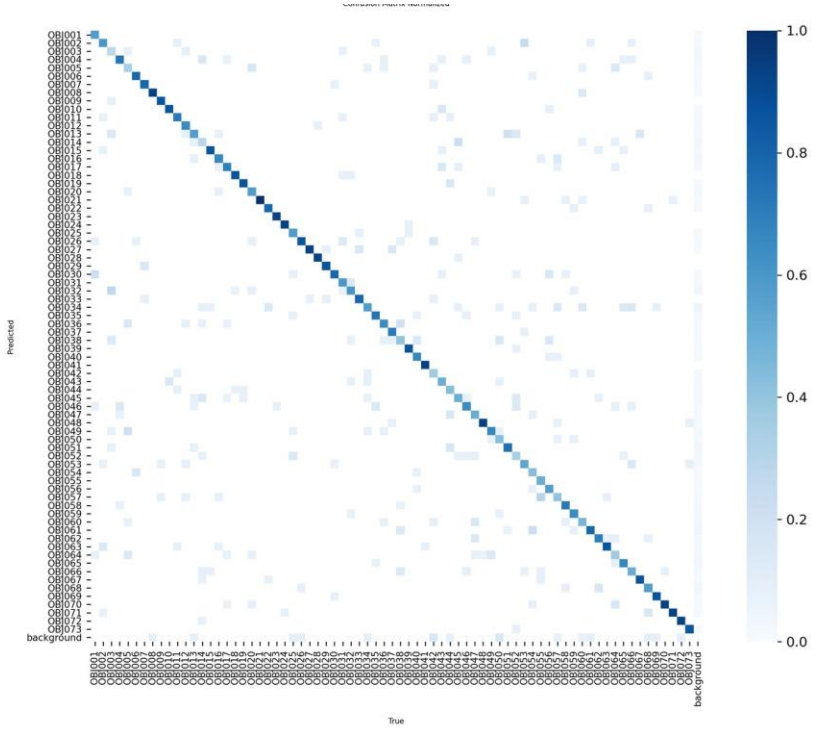


Figure 5. Normalised confusion matrix of the selected model on the main test set (rows: predicted, columns: true; the last row and column are background). Off-diagonal entries concentrate within look-alike groups.

8.6 Qualitative results and the Object ID output

Figures 6 and 7 show detections on test images of every grid size, with true boxes in green and predicted boxes in red. Predicted boxes closely follow the true boxes; visible failures are a missed, very small object in the 5x5 example and low-confidence boxes on small objects. For any input image, the detect step prints and saves (as CSV and JSON) one line per object, as required by the milestone (Table 12).

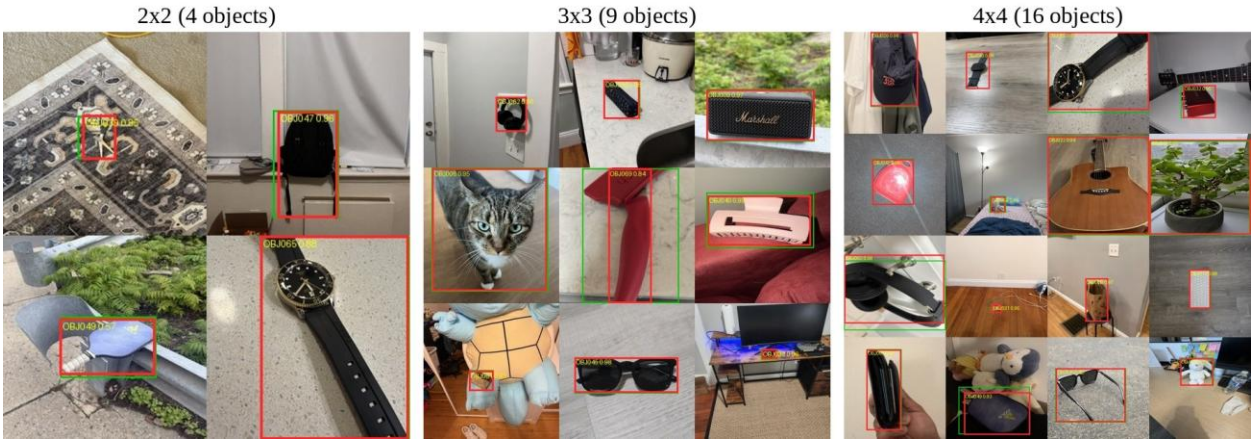


Figure 6. Detections on 2x2, 3x3 and 4x4 test images. Green: true boxes; red: predicted boxes with Object ID and confidence.

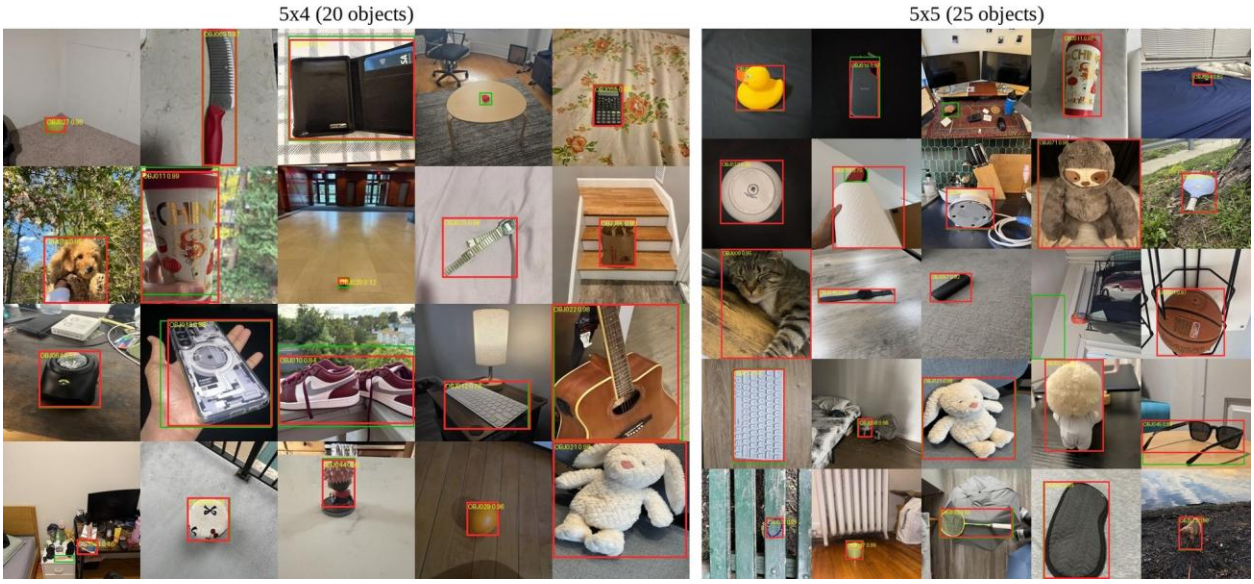


Figure 7. Detections on 5x4 and 5x5 test images (20 and 25 objects).

Table 12. Output of the detect step for the 2x2 test image of Figure 6 (boxes in pixels of the input image, from top-left (x1, y1) to bottom-right (x2, y2)).

Object ID	Confidence	x1	y1	x2	y2
OBJ047	0.95	303	86	370	206
OBJ039	0.26	84	90	123	141
OBJ065	0.97	284	225	448	447
OBJ049	0.39	86	314	169	404

8.7 An interactive explorer

Everything in this report can also be explored in an interactive interface, The EYE (Explainable YOLO Explorer), which runs locally from a single Python file (*m2_eye_app.py*) on the trained detector (Figures 8 and 9).



Figure 8. The EYE after a detection: controls along the top, the image in the pupil with the score ring, and the report card below.

Each part of the interface has one job:

- **Controls (top):** choose the grid size, a section or look-alike mix, image conditions, the seed, the model and the threshold.
- **Left panel:** upload an image, build a random test grid, run a stress test over many grids, or read the model metrics.
- **The eye:** shows the image; its iris is the Detect button and then becomes a score ring with one segment per object.

- **Report card:** the score, every error explained in one sentence with full details, each object next to training photos of its class, and a threshold sweep.
- **Right panel:** download the results (PNG, CSV, JSON), save a snapshot for slides, and switch titles, theme and motion.
- **How was I built?:** the milestone told as a ten-station construction journey, one flip card per step (Figure 9).

The explorer currently runs on the machine that holds the model and the photos; hosting it for use from any system is planned (Section 11).



Figure 9. The 'How was I built?' section of The EYE: the milestone as a ten-station construction journey.

9. Alternative Pipelines Explored by the Group

The group trained eight detectors in total. Runs 1 to 5 belong to the main pipeline described in Sections 4 to 8 and differ only in architecture and fine-tuning (Table 8); run 5 is the selected model. Runs A, B and C are three alternative end-to-end pipelines built in parallel, each combining a different open-vocabulary labelling method with a different detector, and none reviewing boxes by hand. Table 13 places all eight side by side; the alternatives are summarised here to put the design choices in context, not in full detail.

Table 13. All eight runs. Runs 1 to 5 are scored on the human-checked test set; scores marked * are reported by runs A to C on their own, automatically labelled test sets and are not comparable with runs 1 to 5.

Run	Detector	Box labels	Objects	Photos left out	Test mAP@0.5	Test mAP@0.5:0.95
1	YOLOv8-s	Main pipeline: OWLv2, image search and self-trained vote; test set checked by a person	73	4.5% (none above 20%)	0.910	0.846
2	YOLO26-s	Main pipeline	73	4.5%	0.921	0.858
3	YOLOv8-s, fine-tuned	Main pipeline, one label rule corrected	73	4.5%	0.932	0.859
4	YOLO26-s, fine-tuned	Main pipeline, one label rule corrected	73	4.5%	0.931	0.863
5 (selected)	**YOLO26-s, fine-tuned twice**	**Main pipeline, all five label rules corrected**	**73**	**4.5%**	**0.947**	**0.875**
A	YOLOv8-s [3], 80 epochs	Grounding DINO [8] and YOLO-World [15], confirmed by a classifier	73	7.9% (9 objects above 20%, the worst 43%)	0.919*	0.852*
B	YOLOE-26s [16], an open-vocabulary YOLO26	Agreement of YOLO-World and YOLOE	72, with about half the photos per object	about 21% of the photos used	0.947*	0.896*
C	YOLO11-s [17], 40 epochs	Agreement of YOLO-World and YOLOE	38 (one course section)	20.0% (16 objects above 20%, the worst 77%)	0.826*	0.749*

These scores are not comparable with each other. Each alternative was scored on its own test grids, whose boxes came from its own automatic labelling and therefore share the style of its training boxes, and from which the photos that pipeline could not box, often the hardest ones, were absent. The runs also split the photos independently: 399 of the 1,041 test objects were training photos of run B, and 22 of run A.

Cross-evaluation. To compare like with like, every model was scored with the same code on the human-checked test set. For each alternative, two kinds of test object were removed, together with any prediction whose centre fell in their 224-pixel cell: objects of classes it cannot predict, and objects whose photo it had been trained on. The selected model was then scored on exactly the same remaining objects (Table 14, Figure 10). With this code the selected model scores mAP@0.5 = 0.943 and mAP@0.5:0.95 = 0.870 on all 1,041 objects, within 0.5 points of the Ultralytics values in Table 8.

Table 14. Cross-evaluation on the human-checked test set. Each row scores an alternative and the selected model (run 5) on the same unseen objects (mAP50 = mAP@0.5, mAP50-95 = mAP@0.5:0.95); precision, recall and fully correct images use each pipeline's own threshold (0.30, 0.15, 0.25 and 0.10).

Run	Objects	mAP50	mAP50-95	Prec.	Recall	Fully correct	Selected mAP50	Selected mAP50-95	Selected fully correct
A	1,019	0.883	0.782	0.906	0.843	11.1%	**0.944**	**0.872**	**34.7%**
B	628	0.779	0.697	0.838	0.677	5.7%	**0.923**	**0.842**	**38.6%**
C	543	0.819	0.720	0.847	0.783	11.3%	**0.949**	**0.872**	**52.1%**

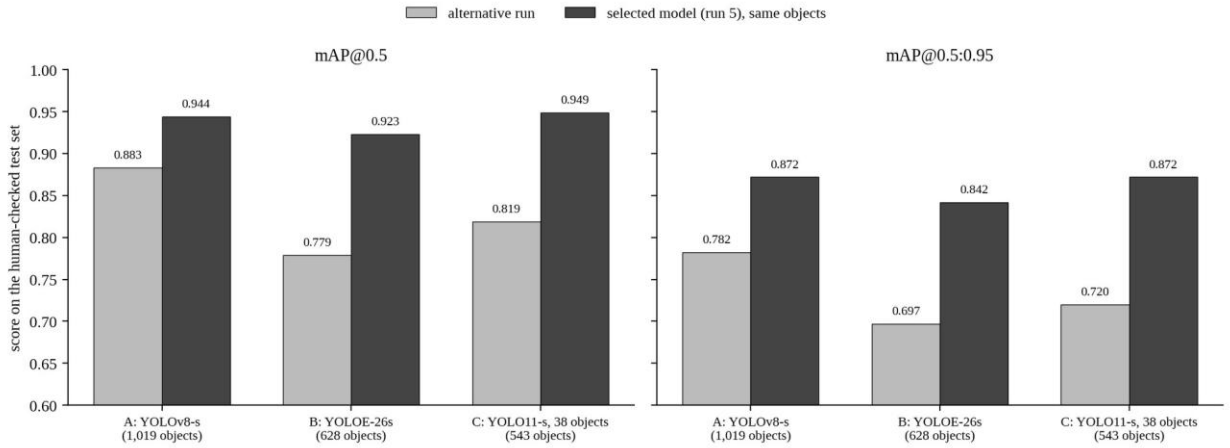


Figure 10. Cross-evaluation on the human-checked test set: each alternative run next to the selected model (run 5) on exactly the same unseen objects (vertical axis from 0.6).

Three conclusions follow, each consistent with the main findings of this report.

1. **Human-checked test boxes change the picture.** Run B matched the selected model's mAP@0.5 on its own test set (0.947) but reached 0.779 on unseen objects with human-checked boxes, a drop of 17 points, and its mAP@0.5:0.95 fell by 20 points. An automatically labelled test set rewards a model for reproducing the labeller's boxes and silently omits the photos the labeller could not handle; a human-checked test set exposes both.
2. **The labels, not the architecture, separated the pipelines.** The four pipelines span three YOLO generations (v8, 11 and 26), yet their ranking follows label quality. The clearest case is the same architecture with different labels: run 1, a YOLOv8-s on the main pipeline's labels, reached 0.910 on the human-checked test set, run A's YOLOv8-s 0.883 on nearly the same objects. The gap is widest on mAP@0.5:0.95 (9 to 15 points), the metric that rewards tight, consistently defined boxes.
3. **The same objects were hard for everyone, and for the reason we found.** The weakest classes of the alternatives are those whose box definitions we corrected (Section 4.6): run A's analog watch (OBJ065, AP@0.5 0.44) and sneakers (OBJ010, 0.63), and the table-tennis racket, the weakest or second-weakest class in runs A and C (0.59 and 0.37). This independently confirms the class-level label audit.

The alternatives also had genuine strengths: run A made only 8 wrong Object IDs on its 1,019 objects (the selected model: 15 on all 1,041 with this code, 16 with the evaluation of Section 7) at its stricter threshold,

at the cost of missing many more objects (recall 0.84 against 0.92), and run B's open-vocabulary detector is a promising starting point that would likely benefit from the same human-checked labels.

Does the cross-evaluation favour the main pipeline? The test set comes from the main pipeline, so the question is fair, and three safeguards answer it. First, the test boxes are not a model's output: every one was checked by a person and 156 were corrected (Table 3), so a model is rewarded for finding the object, not for imitating another model's boxes. Second, each alternative is scored only on objects it never trained on and only on classes it can predict, at its own threshold and with the same code, and the selected model is scored on exactly the same objects. Third, the gap is not a product of the main pipeline's extra fine-tuning: run 1, its first YOLOv8-s without any fine-tuning or label correction, already reaches 0.910 on the full test set, above every alternative (0.779 to 0.883). One residual effect remains: the test boxes follow the main pipeline's box definitions, such as one box around both shoes, so an alternative that defined an object differently loses some overlap on those few classes. The gap, however, persists at mAP@0.5, the metric most tolerant of such differences.

10. Discussion

Labels mattered more than the model. The two architectures differ by about one point of mAP, whereas correcting the definition of single classes changed their AP by up to 0.7 (OBJ051: from 0.30 to 0.995). Method agreement does not guarantee a consistent definition of the object: the text prompt read "tissue" as the paper and the shoes as single shoes, and the self-trained model faithfully learned the same reading. Only plotting every box of a class side by side exposed these errors. We consider this class-level audit the most transferable lesson of the project.

The test set is harder than validation. Test mAP@0.5 is 2.5 points below validation (0.947 against 0.972). Test boxes were all checked or drawn by a person and include the hardest photos, which were fixed by hand; most validation boxes were accepted automatically and share the style of the training boxes. The test results are therefore the conservative and trustworthy measure.

Independent work led to the same conclusion. The three alternative pipelines (Section 9) chose different labellers and detectors, and all were limited by the same few object definitions; scored on human-checked boxes, each fell well below the selected model. Agreement between automatic labellers is therefore not enough: the box definition must be audited, and the test boxes checked by a person.

Where the model fails. Missed objects outnumber wrong Object IDs by more than four to one. The misses are mostly objects that occupy only a few pixels of their 224-pixel photo; the wrong IDs concentrate in look-alike groups. Degraded images remain a separate, much harder condition.

Co-occurring objects. A third source of error lies in how the dataset was photographed. Some objects were photographed in shared settings, so one appears, unlabelled, in the background of another's photos: the apple (OBJ007) and the tissue box (OBJ051), and the tennis ball (OBJ027) and the mouth-guard case (OBJ042), appear in each other's photos. Each photo is labelled with a single object, so during training the second object counts as background, which lowers the model's confidence for its class. At test time a correct detection of the second object counts as an extra box, and the one-box-per-cell rule can keep it instead of the labelled object, turning one correct observation into a miss and an extra. This is a property of the dataset rather than of the model.

10.1 Threats to validity

- **Test-guided audits.** Some label audits (Section 4.6) were prioritised using per-class test results. The corrections applied one fixed rule to all splits and every test box was human-verified regardless, but the choice of *which* classes to audit was informed by test data, so the reported test scores may be slightly optimistic.
- **Synthetic grids.** Each object sits in its own cell with its own background and never overlaps another object. The model is not evaluated on real photographs containing several objects, where accuracy would likely be lower.
- **Automatic training labels.** An estimated 3% of the automatically accepted training boxes are imperfect (97.1% correct in a random sample).
- **Small evaluation samples.** About 14 test photos per class and 13 to 16 images per grid size make per-class and per-size figures uncertain; the bootstrap interval for fully correct images (23.6 to 45.8%) illustrates the size of this uncertainty.
- **Confounded improvements.** The fine-tunes combine data changes with extra training epochs, so the gain attributed to each change is indicative rather than isolated.
- **Cross-evaluation scope.** The alternatives are scored on subsets of the test set (543 to 1,019 objects) after removing photos they trained on, and the test boxes follow the main pipeline's box definitions; differences of a few points between the alternatives are within sampling noise, although the gap to the selected model is large and consistent.

11. Limitations and Future Work

The most promising next steps follow directly from the error analysis. **More photos of look-alike objects**, or a larger model such as YOLO26-m, should reduce the remaining confusions; **stronger degradation training** could recover robustness to low-quality images without the trade-off observed in Section 8.4; and an evaluation on **real photographs of several objects** would measure how far the model transfers beyond synthetic grids. Reviewing all automatically accepted training boxes by hand would remove the estimated 3% label noise, but at a high cost for an expected gain of less than one point. Labelling the second object in photos where two course objects appear together would remove the co-occurrence errors of Section 10. Finally, we plan to host the interactive explorer of Section 8.7 as a web service, so that it can be used from any browser on any system without installation; because it shows photos from the course dataset, access will be restricted to the course unless permission is given to share it more widely.

12. Conclusion

We built a detector that reports the Object ID and bounding box of every object in grids of 2x2 to 5x5 photos and in single photos from the course dataset. Because no boxes existed, the main effort went into labels: a consensus of two automatic box sources and a self-trained second opinion, human verification of every test box, measured accuracy for automatically accepted boxes, and a class-level audit that corrected five inconsistent object definitions. A fine-tuned YOLO26-s, selected on validation data, reaches mAP@0.5 = 0.947 and mAP@0.5:0.95 = 0.875 on human-verified test boxes, detects single photos without error and rarely assigns a wrong Object ID. Cross-evaluated on the same unseen, human-checked objects, it

outperforms three alternative pipelines built within the group by 6 to 14 points of mAP@0.5, which confirms that trustworthy labels, rather than the choice of architecture, determine detection quality on this dataset. Its remaining weaknesses are visually similar objects, very small objects and degraded images. An interactive explorer built on the detector makes every result, error and design step inspectable (Section 8.7).

Code: The Github repository contains code only: the box pipeline (*M2_Boxes.ipynb*) and the detection pipeline (*M2_Detect.ipynb*). Images, grids and labels are kept out of the repository, as required.

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Appendix A. Photos Used and Left Out per Object ID

Object ID	Used / photos	Left out	Object ID	Used / photos	Left out	Object ID	Used / photos	Left out
OBJ001	95/95	0.0%	OBJ026	94/95	1.1%	OBJ051	105/105	0.0%
OBJ002	95/95	0.0%	OBJ027	87/95	8.4%	OBJ052	92/95	3.2%
OBJ003	95/95	0.0%	OBJ028	95/95	0.0%	OBJ053	85/90	5.6%
OBJ004	95/100	5.0%	OBJ029	93/94	1.1%	OBJ054	93/95	2.1%
OBJ005	93/99	6.1%	OBJ030	95/95	0.0%	OBJ055	93/95	2.1%
OBJ006	87/95	8.4%	OBJ031	91/95	4.2%	OBJ056	94/95	1.1%
OBJ007	95/95	0.0%	OBJ032	95/95	0.0%	OBJ057	86/95	9.5%
OBJ008	94/95	1.1%	OBJ033	93/95	2.1%	OBJ058	88/95	7.4%
OBJ009	92/95	3.2%	OBJ034	95/95	0.0%	OBJ059	76/95	20.0%
OBJ010	95/95	0.0%	OBJ035	95/95	0.0%	OBJ060	95/95	0.0%
OBJ011	91/93	2.2%	OBJ036	88/95	7.4%	OBJ061	95/95	0.0%
OBJ012	86/95	9.5%	OBJ037	92/95	3.2%	OBJ062	75/91	17.6%
OBJ013	78/94	17.0%	OBJ038	86/95	9.5%	OBJ063	90/94	4.3%
OBJ014	79/93	15.1%	OBJ039	94/95	1.1%	OBJ064	80/95	15.8%
OBJ015	78/95	17.9%	OBJ040	95/95	0.0%	OBJ065	95/95	0.0%
OBJ016	92/95	3.2%	OBJ041	89/95	6.3%	OBJ066	94/94	0.0%
OBJ017	105/106	0.9%	OBJ042	81/95	14.7%	OBJ067	95/95	0.0%
OBJ018	95/95	0.0%	OBJ043	95/95	0.0%	OBJ068	95/95	0.0%
OBJ019	79/95	16.8%	OBJ044	78/95	17.9%	OBJ069	78/93	16.1%
OBJ020	95/95	0.0%	OBJ045	94/95	1.1%	OBJ070	95/95	0.0%
OBJ021	95/95	0.0%	OBJ046	95/95	0.0%	OBJ071	79/95	16.8%
OBJ022	93/94	1.1%	OBJ047	93/95	2.1%	OBJ072	94/95	1.1%
OBJ023	84/89	5.6%	OBJ048	95/95	0.0%	OBJ073	95/95	0.0%
OBJ024	95/95	0.0%	OBJ049	97/99	2.0%			
OBJ025	87/94	7.4%	OBJ050	88/95	7.4%			

Left out: 314 of 6,942 photos (4.5%): 6 not photos of the object (OBJ054_017 and five OBJ004 background shots) and 308 without a reliable box (the automatic methods disagreed and the photo was not reviewed). No Object ID exceeds 20%.

Appendix B. Reproducibility

Hardware and software: NVIDIA RTX 5070 Laptop GPU (8 GB), PyTorch 2.11 (CUDA 12.8), Ultralytics 8.4, Hugging Face Transformers 5.2; random seed 42. Code: *M2_Boxes.ipynb* (box pipeline and review tool), *M2_Detect.ipynb* (grids, training, evaluation, detection) *m2_cross_eval.py* (the cross-evaluation of Section 9) and *m2_eye_app.py* (the interactive explorer of Section 8.7); every step logs, checkpoints and resumes, and every component was tested on synthetic data with known answers.